BMS Slave

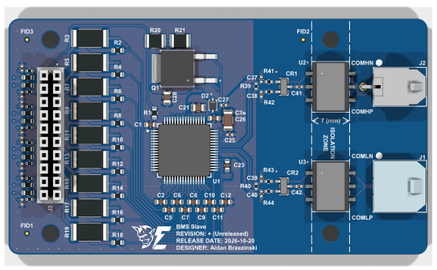
Variant: CHECKED

2026-10-20

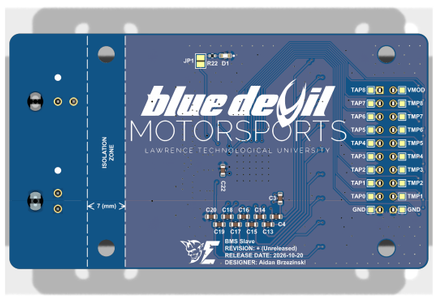
Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11	.....	21	.....	31	.....
2	Block Diagram	12	.....	22	.....	32	.....
3	Project Architecture	13	.....	23	.....	33	.....
4	Connectors	14	.....	24	.....	34	.....
5	Cell Sense & Balance	15	.....	25	.....	35	.....
6	Power & Reference	16	.....	26	.....	36	.....
7	Temperature Sense	17	.....	27	.....	37	.....
8	Daisy Chain	18	.....	28	.....	38	.....
9	Mechanical	19	.....	29	.....	39	.....
10	Revision History	20	.....	30	.....	40	.....

TOP VIEW



BOTTOM VIEW



NOTES

TI BQ79616PAPRQ1 Datasheet

Not fitted components are marked as ✗

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

DESIGN NOTE: Example text for cautionary design notes.

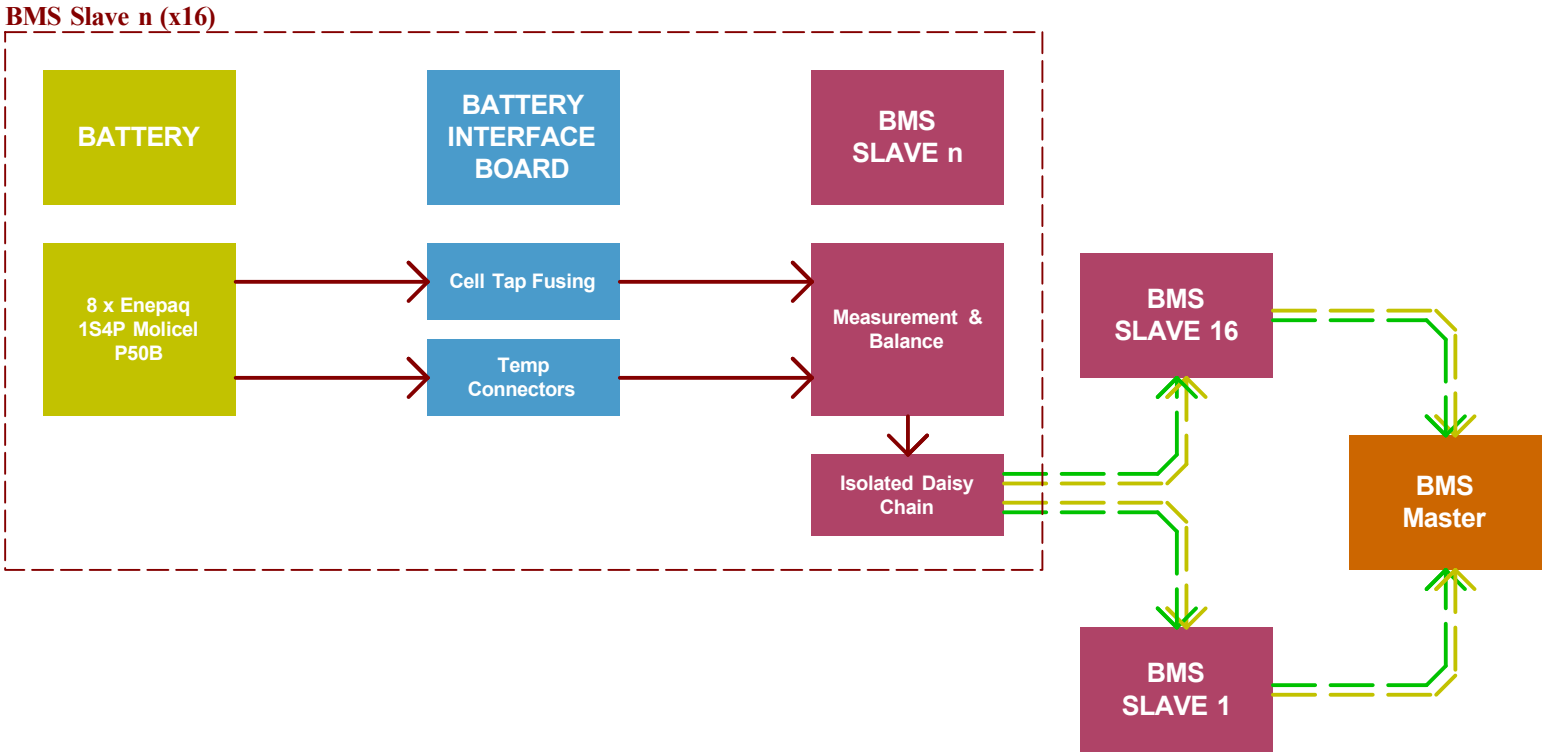
DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

blue devil MOTORSPORTS

Comments:	Company: Blue Devil Motor Sports			Variant:
	Board Name: BMS_SLAVE			Project Name: BMS Slave
Sheet Title: bms_slave	File Name: bms_slave.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 10

[2] Block Diagram

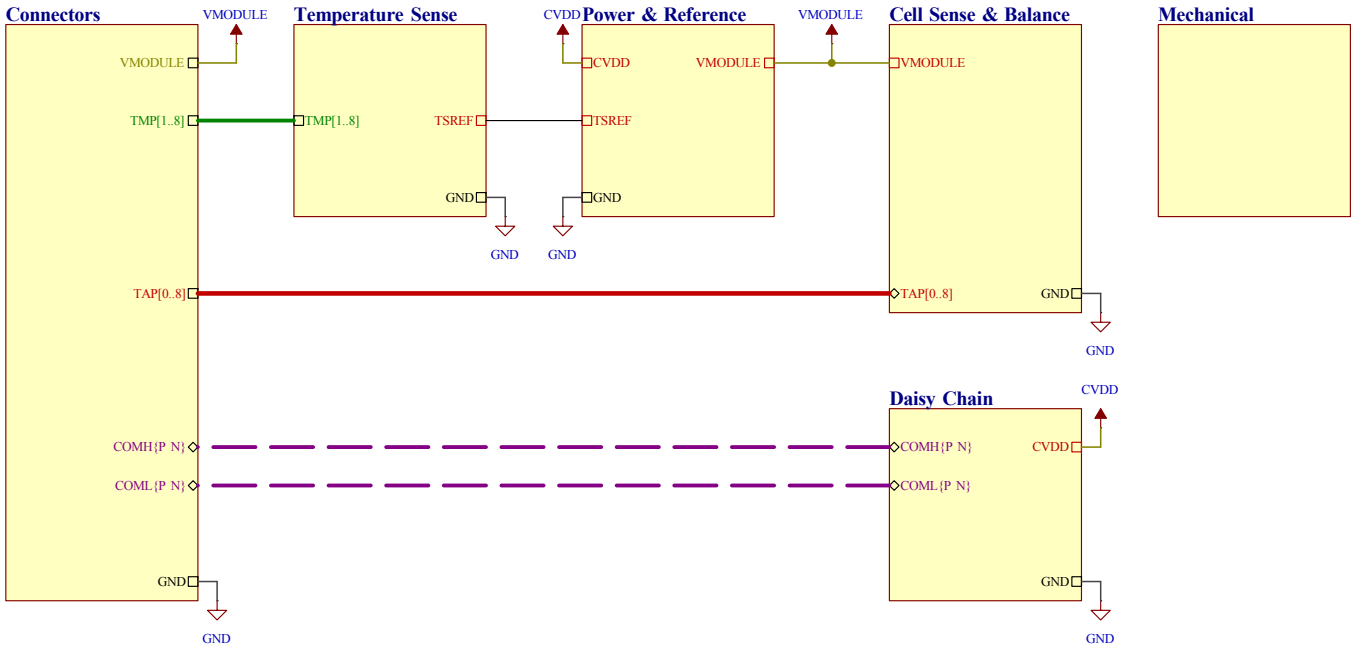


Target specifications:

Input voltage:	23.2 - 33.6 V
Balancing	190mA passive
Temperature	8 x Enepaq sen.
Isolated Com.	600V

Comments:	Company: Blue Devil Motor Sports		Variant:	
	Board Name: BMS_SLAVE		Project Name: BMS Slave	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: A4	Sheet: 2 of 10

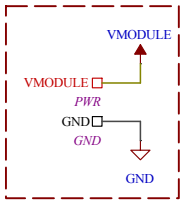
[3] Project Architecture



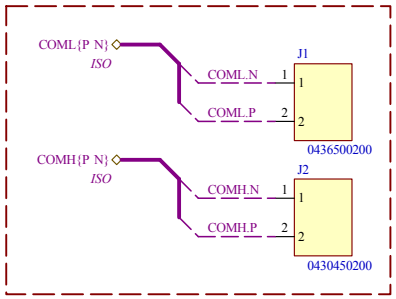
Comments:	Company: **Blue Devil Motor Sports**		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: **A4**	Sheet: **3** of **10**

[4] Connectors

Power

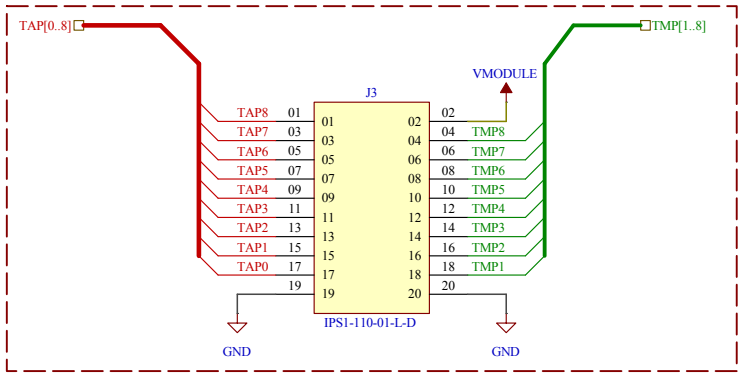


BMS Comminication Connectors



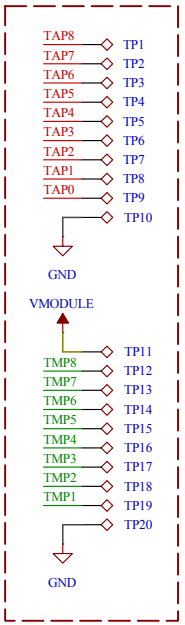
LAYOUT NOTE: CONNECTORS
1. J1/2 (daisy) at board edge beside their transformers.
2. Ensure that board to board connector matches battery interface board.
3. Tab fuses at CELL END of tap harness, >= 67.2 V rated (FSAE EV.8.4.3), In this case on the battery interface board.

Board to Board Connector



DESIGN NOTE:
Using a single ground on connector assumes that on the battery interface board there is going to be a common ground for all temp wires. Need to route to correct ground plane for noise.

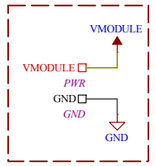
Test Points



Comments:	Company:		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Connectors	File Name: connectors.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: **A4**	Sheet: **4** of **10**

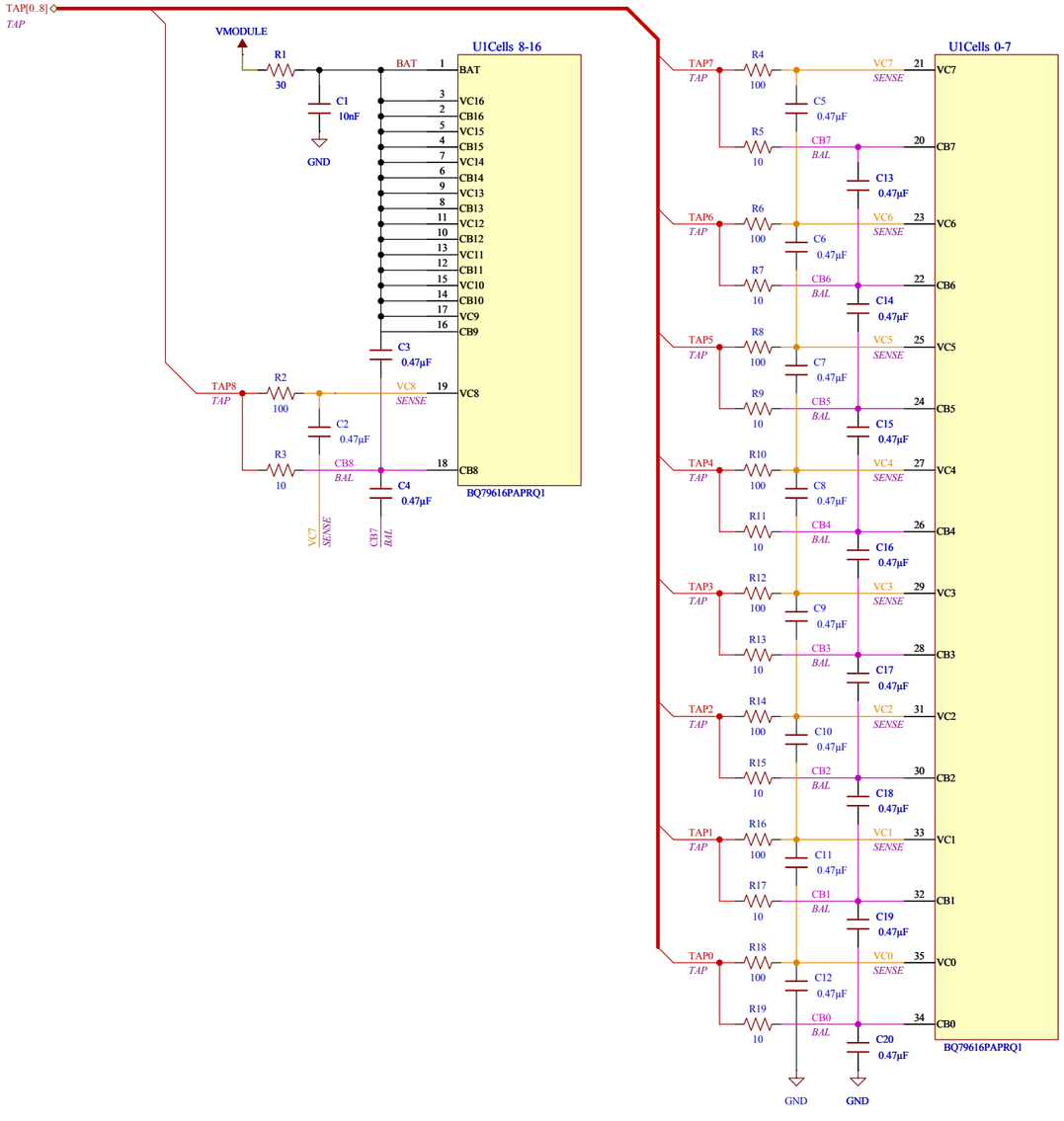
[5] Cell Sense & Balance

Power



LAYOUT NOTE: CELL SENSE (SLUSE81F)
1. VC/CB filter caps at U1 pins; RC values per 8.2.1.2.1 (p.192).
2. VC and CB traces run in parallel, impedance matched (within 10mm based off TI example layout no other spec listed); CB traces sized for ~190 mA balancing (10.1.3, p.205).
3. Unused VC9-16/CB9-16 tie to BAT pin with minimum trace; never to a wire or cable (Fig. 8-2a, p.192).
5. Balancing heat: ~3 W/board at 4 channels; spread R_CB (2512).

Cell Sense & Balance

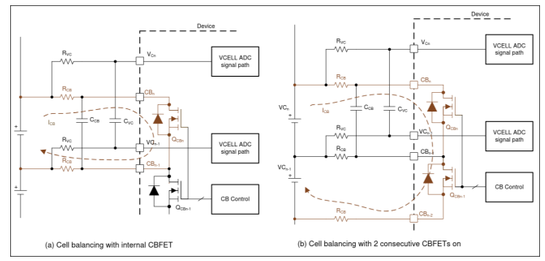


DESIGN NOTE: VC FILTER (R_VC / C_VC)
(TI SLUSE81F 8.2.1.2.1, p.192)
1. R_VC = 100 ohm, C_VC = 0.47 uF. TI fixed values - do not change.
2. Differential RC: cap across adjacent VC lines.
   fc ~ 1/(2*pi*200*0.47u) = 1.7 kHz.
3. Values also give hot-plug protection when harness is connected live.
4. 100 ohm keeps input-current error < 1 mV.
5. Cap sees 1 cell max (4.2 V). TI min 16 V; use 25 V X7R (DC-bias margin).

DESIGN NOTE: CB BALANCE RESISTORS (R_CB = 10 ohm, 2512 / 1 W)
(TI SLUSE81F 7.3.3 p.36; Rdson 6.5 p.13; 240 mA limit p.1)
1. I_CB = Vcell / (2*R_CB + Rdson). Current flows through BOTH R_CB (CBn and CBn-1) and the internal balance FET.
2. Rdson = 1.45-4.6 ohm (rises with die temp, part-to-part; spec Vcell > 2.8 V).

Vcell	I_CB (Rdson 1.45 / 4.6)	P per R_CB	P per cell
3.0 V	140 / 122 mA	0.15-0.20 W	~0.4 W
3.6 V	168 / 146 mA	0.21-0.28 W	~0.6 W
4.2 V	196 / 171 mA	0.29-0.38 W	~0.8 W

3. Max 196 mA < 240 mA limit (240 mA rated at 75 C ambient).
4. ~1 % SOC/hr on 20 Ah Enepaq block.
5. Less heat option: R_CB = 20 ohm -> ~100 mA, ~1/4 power.



DESIGN NOTE: CB FILTER CAPS + BAT FILTER
(TI SLUSE81F 8.2.1.2.1-8.2.1.2.2, pp.192-193)
1. C_CB = 0.47 uF (TI fixed): hot-plug + filtering, fc ~ 17 kHz.
2. Clean CB path needed for VC-vs-CB (AUX ADC) diagnostic comparison.
3. BAT: R1 = 30 ohm, C1 = 10 nF (TI fixed, hot-plug).

Comments:	Company:		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Cell Sense & Balance	File Name: cell_sense_and_balance.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Cell Sense & Balance/		Reviewer:	Size: **A3**	Sheet: **5** of **10**

[6] Power & Reference

LAYOUT NOTE: U1 GROUNDING (TI SLUSE81F 10.1.1)

All grounds are one net (GND) in the schematic. In layout, each bypass cap returns to its own ground pin through a small L1 copper island. Islands join the L2 GND plane through vias only. Do NOT connect islands to each other on L1.

LAYOUT NOTE: RULES

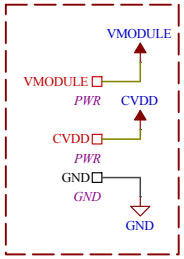
1. Cap within 2 mm of its pin, same layer, no via between cap and pin.
2. Cap ground pad -> its island -> via to L2. Not a shared trace.
3. Keep pins 40-43 (COMH/COML) clear; place caps above and below the daisy-chain exit, not in it.
4. Thermal pad (65): 36 vias to L2 GND plane, tented/plugged.

GROUND	PIN	CAPS RETURNING TO IT
AVSS	39	AVDD (38), BAT filter, VC0 filter, CB0 filter, thermal pad, (65)GPIO filter caps.
DVSS	50	DVDD (49)
CVSS	46	LDOIN (47), NEG5V (44), CVDD (45), TSREF (51), daisy-chain caps.
REFHM	36	REFHP (37) ONLY - keep separate from AVSS on L1, join at L2 plane

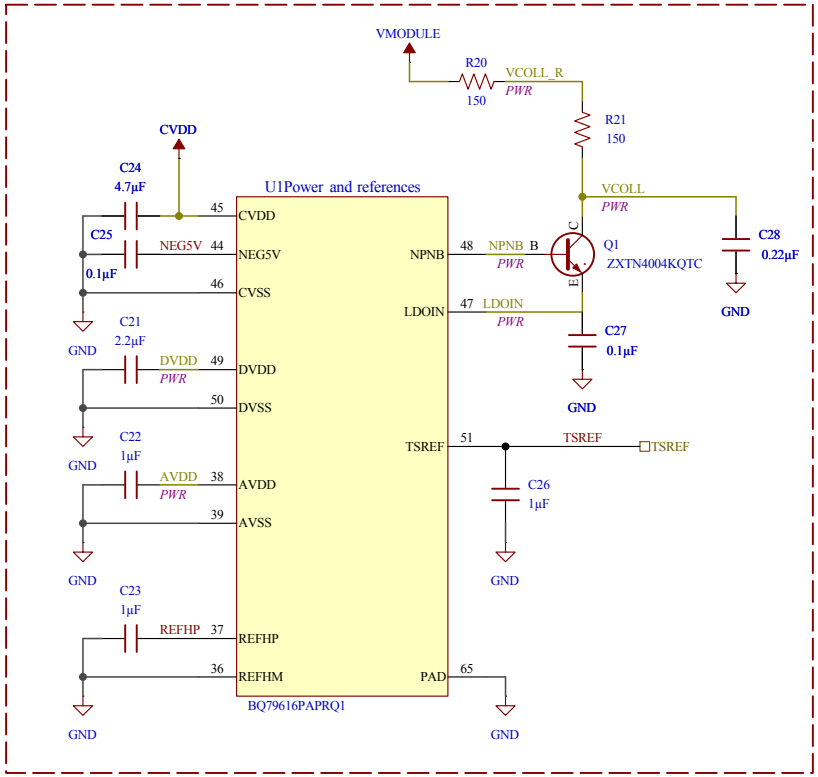
DESIGN NOTE: CVDD LOAD BUDGET
(TI SLUSE81F 7.3.1.5 p.23; ILIM 6.5 p.12; ISHDN 6.5 p.11; modes 7.4 p.103)

1. CVDD = 5 V LDO for daisy chain + I/O. Allowed external load:
   10 mA in ACTIVE/SLEEP, 5 mA in SHUTDOWN. Current limit 35 mA min.
2. External loads on this board:
   RX tie to CVDD ~0 mA (input leakage only)
   LED D1 + R22 2k0 ~1.5 mA (only when JP1 bridged)
   TOTAL ~1.5 mA of 10 mA -> OK
3. Daisy-chain comm current (~10 mA during broadcast) is internal, not part of the 10 mA external budget.
4. DO NOT add other loads to CVDD. Temp sensors run from TSREF; balancing current comes from the cells.
5. SHUTDOWN: CVDD regulator is off, but the pin follows LDOIN (for WAKE detection) - it is NOT 0 V. IC draws 16-23 uA in SHUTDOWN; LED would add ~1 mA (~50x). JP1 OPEN in the car. Bridge only for bench debug.

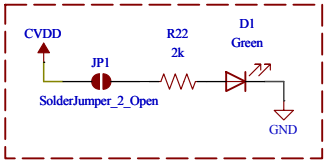
Power



Power and Reference



LED Indicator



LAYOUT NOTE: POWER (TI SLUSE81F)

1. Bypass caps as close as possible, REFHP most critical (10.1.2, p.205).
2. Ground groups per table; islands on L1, join on L2 (10.1.1, Fig. 10-1, p.203-204). NEG5V -> CVSS per 10.1.1 (Fig. 8-1 shows AVSS; same net).
3. Q1 DPAK: >= 4 cm2 VCOLL copper on L1+L4, stitched; away from U1 and R_CB. R20/R21 share dissipation (8.2.1.2.2, Fig. 8-3, p.193).
4. Thermal pad: PAP0064F land pattern, vias tented/plugged (pkg drawing notes 8-9, p.220).

DESIGN NOTE: CVDD LOAD BUDGET
(TI SLUSE81F 7.3.1.5 p.23; ILIM 6.5 p.12; ISHDN 6.5 p.11; modes 7.4 p.103)

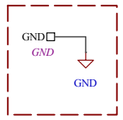
1. CVDD = 5 V LDO for daisy chain + I/O. Allowed external load:
   10 mA in ACTIVE/SLEEP, 5 mA in SHUTDOWN. Current limit 35 mA min.
2. External loads on this board:
   RX tie to CVDD ~0 mA (input leakage only)
   LED D1 + R22 2k0 ~1.5 mA (only when JP1 bridged)
   TOTAL ~1.5 mA of 10 mA -> OK
3. Daisy-chain comm current (~10 mA during broadcast) is internal, not part of the 10 mA external budget.
4. DO NOT add other loads to CVDD. Temp sensors run from TSREF; balancing current comes from the cells.
5. SHUTDOWN: CVDD regulator is off, but the pin follows LDOIN (for WAKE detection) - it is NOT 0 V. IC draws 16-23 uA in SHUTDOWN; LED would add ~1 mA (~50x). JP1 OPEN in the car. Bridge only for bench debug.

blue devil MOTORSPORTS

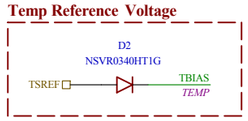
Comments:	Company:		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Power & Reference	File Name: power_and_reference.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Power & Reference/		Reviewer:	Size: **A4**	Sheet: **6** of **10**

[7] Temperature Sense

Power



Temp Reference Voltage

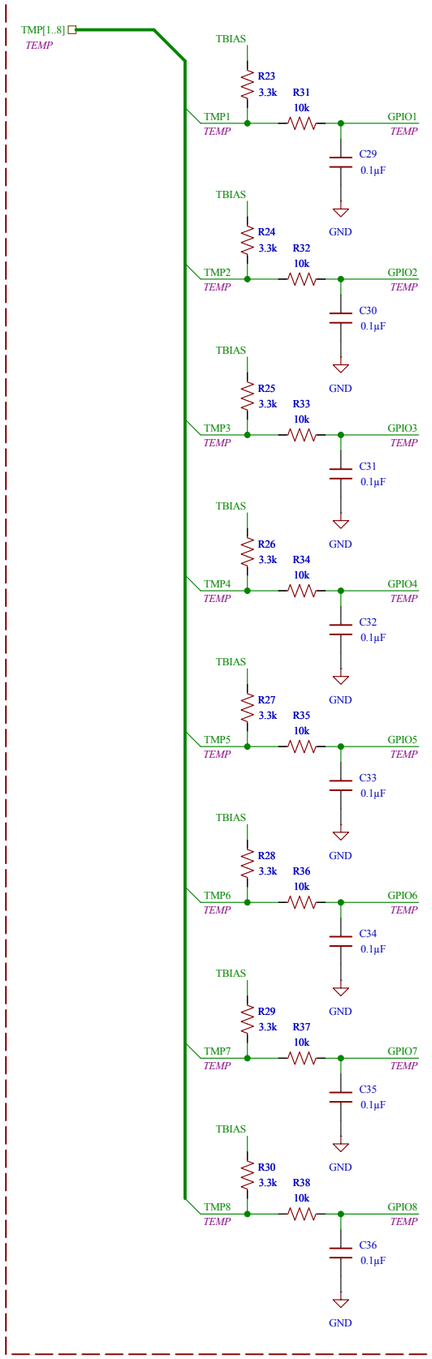


D2: blocks back-feed into TSREF if a TMP wire shorts to a cell tap. ~0.3 V drop is harmless withshunt-reference sensors.

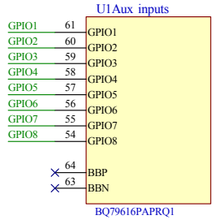
DESIGN NOTE: BQ7961x COMPARATOR SETTINGS (SLUSE81F 7.3.x)

1. OT comparator (10-39 % of TSREF) works with these sensors:
OT_THR 31 % = 1.55 V ≈ 55°C (margin under 60°C); 30 % ≈ 61°C.
Hysteresis 2 % ≈ 0.1 V.
2. UT comparator (66-80 %) cannot detect cold (sensor max 2.44 V = 49 %). Use it as an OPEN-WIRE detector: an open sensor pulls GPIO to TBIAS (≈94 %) and trips UT. A shorted sensor trips OT.
3. OTCB (10-24 % = 0.5-1.2 V) can never trigger (sensor floor 1.30 V). Balancing thermal limits are firmware + die temp only.

Sensor Bias & RC Filter (x8)



Aux Inputs



LAYOUT NOTE: TEMPERATURE SENSE

1. Place R23-R38 and C29-C36 at J3. Filter caps return to GND beside J3 pins 19/20 so harness noise goes straight back out the connector, not under U1.
2. Sensor returns share J3-19/20 with U1's supply return. Never join them to TAP0 (J3-17) on either board.
3. Filtered GPIO1-8 traces may run the length of the board; keep them over the GND plane, away from COMH/COML.
4. Add 2-3 GND vias at J3-19/20.
5. TBIAS trace carries ≈8 mA DC; 0.2 mm min width.

DESIGN NOTE: ENEPAQ TEMPERATURE SENSORS

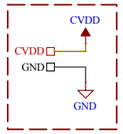
1. Each Enepaq 1x4 module has one 2-wire sensor output (hottest of its sensors, analog OR'd). The sensor is a shunt voltage reference ("zener-like"), NOT an NTC: Vout is set by temperature, not by bias current.
2. Vout vs temp (Enepaq Table 4): -40°C 2.44 V | 25°C 1.86 V | 60°C 1.51 V | 120°C 1.30 V. Use linear interpolation.
3. Bias: TBIAS (≈4.7 V) through 3.3 kΩ gives 0.7-1.0 mA per sensor(spec 0.4-15 mA). 8 sensors ≈ 8 mA from TSREF (limit ≥15 mA).
4. Read the absolute GPIO voltage and convert with Table 4. Do not use TI's NTC ratiometric formula. D2's drop only changes bias current, not the reading.
5. Sensor is isolated from the cells up to 60 V. Fine at 8S (33.6 V max); re-check if module count per slave grows.
6. R31-R38 / C29-C36: 10 kΩ / 0.1 µF filter (fc ≈ 160 Hz). The 10 kΩ also limits GPIO clamp current on a harness fault.

blue devil MOTORSPORTS

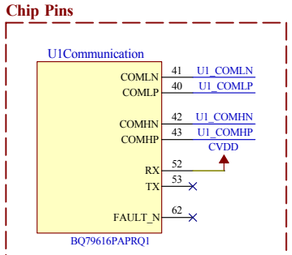
Comments:	Company:		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Temperature Sense	File Name: temperature_sense.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Temperature Sense/		Reviewer:	Size: **A3**	Sheet: **7** of **10**

[8] Daisy Chain

Power



Chip Pins



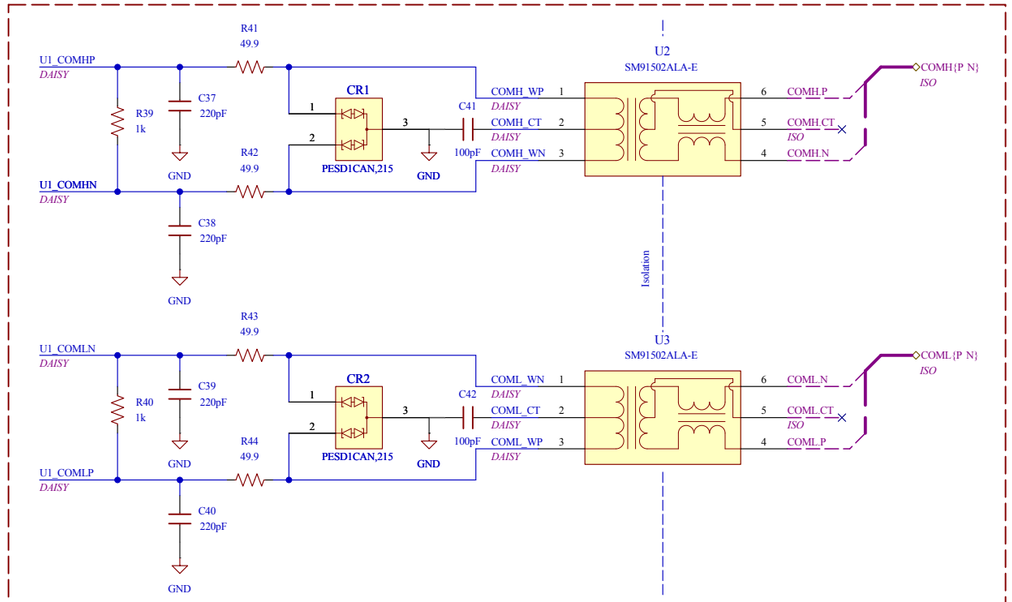
DESIGN NOTE: DAISY CHAIN ISOLATION

1. Transformer isolation per SLUSE81F Table 8-5 / Fig 8-9: R_TERM 1 kΩ, R_SERIES 49.9 Ω, C_BYPASS 220 pF/50 V C0G, centre-tap cap 100 pF to GND, TVS PESD1CAN on chip side.
2. Cable-side centre taps (U2-5, U3-5) are left NC per Fig 8-9.
3. Stack device: RX tied to CVDD, TX and NFAULT left NC (SLUSE81F 8.2.1.2.6).
4. U3 is drawn with chip-side pins 1/3 opposite to U2. Polarity is preserved end to end (P->P, N->N).
5. Ring architecture: COMH (J2) goes up the stack, COML (J1) goes down. J1 = 43650-0200 single-row, J2 = 43045-0200 dual-row so a cable cannot connect COMH to COMH.
6. Use the same transformer PN on both ends of every link, including the master (TI SLVAEP4 5.1).

LAYOUT NOTE: DAISY CHAIN (TI SLUSE81F 10.1.4, Fig 10-3)

1. COMH and COML pairs: short, straight, same layer, parallel, length matched. No vias in the pairs.
2. R_TERM and C_BYPASS close to U1 pins 40-43.
3. Solid GND under the chip-side parts (R, C, TVS) is required for the bypass and TVS returns. Keep unrelated traces out of this area, and never route a COM pair across a plane seam.
4. No copper on ANY layer under the transformer bodies or between the transformer cable side and J1/J2.
5. Transformer barrier: 7.11 mm pad gap meets EV.7.5.7 300-600 V coated creepage (4.0 mm) and clearance (3.0 mm). Keep >= 4.0 mm on all layers. No plated holes, vias or pours inside the isolation zone.
6. Place each transformer next to its connector.
7. TP11-TP14: bare SMD pads on the trace, no stubs.

Communication Isolation



CAUTION: ISOLATION SPACING (FSAE 2027 EV.8.3.2, EV.7.5.7)

1. EV.8.3.2: the BMS must have galvanic isolation at every Module-to-Module boundary. U2/U3 provide it. All slaves are identical, so every barrier is designed for the worst case: full pack voltage, <128S x 4.2 V = 537.6 V> max (EV.3.3.2 limit: 600 V DC).
2. EV.7.5.7 spacing, 300-600 V DC row:
Clearance (through air): 3.0 mm
Creepage, general (uncoated): 20.0 mm
Creepage, under conformal coating: 4.0 mm
3. This board IS conformally coated, so the 4.0 mm creepage and 3.0 mm clearance apply. Uncoated it would NOT comply (7.11 mm < 20 mm).
4. Rule applies on the same layer AND between layers. Keep >= 4.0 mm on every layer between chip-side copper (inner GND pours, vias, TVS / CT-cap pads) and cable-side copper (U2/U3 pins 4-6, J1/J2). Actual: 7.11 mm transformer pad gap (SM91502ALA land pattern).
5. No plated holes, vias or copper pours inside the isolation zone.
6. Mark the zone on top AND bottom silkscreen (boundary lines + "ISOLATION ZONE"), per EV.7.5.6 / EV.7.5.7a.
7. U2/U3 isolation rating and the daisy-chain cable insulation must be rated >= max TS voltage (EV.5.2.1, EV.7.3.2).
8. Keep a spare coated board or photos for inspection (EV.7.5.8).

Comments:	Company:		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Daisy Chain	File Name: daisy_chain.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Daisy Chain/		Reviewer:	Size: **A3**	Sheet: **8** of **10**

[9] Mechanical

DESIGN NOTE: FIDUCIALS / LOGOS / MARKING

1. FID1-FID3: 1.0 mm copper dot, 2.0 mm mask opening and keep-out, three non-collinear points near the board corners.
2. Logos are silkscreen graphics only: set "Exclude from BOM" and "Exclude from simulation". Keep them off pads and vias, and out of the isolation zone.
3. Mark the isolation zone on top AND bottom silk (boundary lines + "ISOLATION ZONE") per EV.7.5.6 / EV.7.5.7a.
4. Board info text uses '${REVISION}' and '${ISSUE_DATE}' so the silkscreen always matches the schematic.

FAB / ASSEMBLY NOTES

1. <4>-layer, <1.6> mm FR-4 (Tg >= <150> °C), <1> oz outer / <1> oz inner.
2. All vias under U1 tented or plugged, including the 36 thermal-pad vias.
3. Copper to board edge >= <0.5> mm on all layers. Coating thins at edges, so don't rely on it there.
4. CONFORMAL COAT REQUIRED (EV.7.5.7 creepage depends on it): <acrylic / urethane / silicone>, rated >= 90 °C (EV.7.2.1), with UV tracer. Both sides; fully cover the isolation zone and U2/U3 pins. Mask J1-J3, test-point pads and mounting holes. Inspect coverage under UV before installation.
5. Build one spare coated board (or keep photos) for technical inspection (EV.7.5.8).
6. JP1 (LED enable) left OPEN for vehicle use.

Mounting Holes

H1 MountingHole
H3 MountingHole
H2 MountingHole
H4 MountingHole

Fiducials

FID1 Fiducial
FID3 Fiducial
FID2 Fiducial

Logos

SAE Electric logo

BMD Logo

LAYOUT NOTE: MOUNTING (FSAE 2027 EV.5.2.2, EV.7.5.2, EV.7.5.6)

1. H1-H6: M3, 3.2 mm NPTH (no plating, no pad, no annular ring).
2. Copper keep-out Ø <8> mm around each hole on ALL layers(M3 washer OD 7 mm + margin).
3. No mounting holes inside the daisy-chain isolation zone. Holes outside the zone must still keep >= 4.0 mm to any cable-side copper (EV.7.5.7, 300-600 V, coated).
4. Board GND = module negative = tractive-system potential. No connection to the chassis or any conductive surface (EV.7.5.2). Mount only with nylon screws, standoffs and washers.
5. If the accumulator container is conductive, mount the board on an insulating plate or bracket rated >= max TS voltage and >= 90 °C (EV.5.2.2a, EV.7.2.1). Never mount directly to the container wall.
6. Positively restrain the board and harnesses so spacing is kept under vibration (EV.7.5.6).

Comments:	Company:		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Mechanical	File Name: mechanical.kicad_sch	Designer: Aidan Brzezinski	Date:	Revision:
Sheet Path: /Project Architecture/Mechanical/		Reviewer:	Size: **A4**	Sheet: **9** of **10**

[10] Revision History

Comments:	Company: **Blue Devil Motor Sports**		Variant:	
	Board Name: **BMS_SLAVE**		Project Name: **BMS Slave**	
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Aidan Brzezinski	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Revision History/		Reviewer:	Size: **A4**	Sheet: **10** of **10**